

The Residue Reads the Constraint: Channel- and Depth-Structured Cross-Round Residue in a Maximal-Wash Construction

The Residue-Trace Engines (28, 29, 29B)

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Abstract

The constraint-priority thesis of the NEXUS program holds that an object is the settled residue of constraint, and that a construction engineered to maximize the wash — perfect local cancellation, all information present at every readout, net signal zero — does not destroy its recoverable structure but displaces it into coupling, where no single readout can see it. This paper tests that claim where it can be checked against ground truth: the synthetic, perfectly rigid environment of SHA-256. A prior result (Engine 27) established that every marginal of the round function is featureless — the working-register phase is uniform, the echo/subdivision phase is a single broad concentration, and the cumulative phase precesses freely with no quantization. Here we measure the residue the only honest way the thesis permits: as cross-round coupling, against a matched null that preserves every per-round marginal exactly and destroys only the coupling between rounds. The coupling is real and sharp. Under a parity readout, the pairwise mutual information between rounds exceeds the matched null by 2.66 bits at lag one, decaying smoothly to zero by lag four — a three-round envelope set by the round function's local diffusion depth. We then exhaust the probe, because a bound is only worth publishing once the lens is ruled out: richer readouts (low-order state bits, an injection-site state shadow) show no coupling at any lag, killing the reading that the residue is a recoverable trajectory, while an explicit carry-

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bit readout reveals a second, distinct structure — silent at lags one through three, then a single sharp echo at lag four (excess 0.12 bits, $z = 366$), then silent again. A dedicated mechanism test (Engine 29B) sources this echo: it is present with equal strength in the pre-recurrence rounds where the message schedule is the raw input and the recurrence is inactive, proving the echo is not the schedule but the register-pipeline latency of the round function's own eight-slot shift. The residue is therefore neither a fog that fades with distance nor a trajectory that can be retraced; it is two distinct constraints in two distinct channels at two distinct depths — a three-round parity envelope and a lag-four carry echo — each invisible to marginals and to richer state shadows, each mechanistically sourced. This confirms the central prediction of constraint-priority while bounding it precisely, and it corrects, with measurement, the proposal that one may relax such a construction and read its full history from the relaxation: the residue reads the constraint within a known, narrow, channel-specific window, and not one step further.

1. The Question the Marginals Forced

In the constraint-first reading, an object is not a thing that sits in a field and obeys forces; it is the local settled residue of constraint, the configuration in which the surrounding tensions cancel without collapsing to undifferentiated sameness. A protein does not fold because constraint pushes it; the folded protein is the visible fold constraint leaves behind. A cryptographic digest is the same kind of object — a terminal knot in a bit-field, the settled result of a fixed sequence of constraints applied to a seed. The program's prior paper established that this reading is not a manner of speaking by measuring three of its consequences directly: the substrate's interior carries no free bits, the dual-wave tension is conserved in every coordinate, and coherence is a boundary service guarded by a conserved charge.

That paper also reported a sustained negative that sets up this one. The hypothesis that the SHA-256 phase trajectory quantizes on the harmonic angle $H = \pi/9$ was tested under three separate canonical extractions and was not supported at word scale: the normalized working-register phase is indistinguishable from uniform, the echo/subdivision split is a single broad concentration at 0.80 radians with no lattice of peaks, and the nine-round cumulative phase precesses freely with a mean distance to closure of 0.78 radians. Under the container ontology this looks like a failed search for a signal presumed to be elsewhere. Under constraint-priority it is informative in a specific way: a construction built to maximize the wash drives every marginal to featurelessness by design — the designers' success at diffusion and the wash are the same fact — and so the recoverable structure, which the construction's bijective round maps preserve rather than destroy, must have been displaced into the coupling between rounds, exactly where a marginal probe is by construction blind.

This paper measures that coupling. The thesis makes a sharp, falsifiable prediction: the displaced tension lives in the cross-round correlation of the trajectory, not in any round's marginal, and it should be detectable as mutual information between rounds that a matched null — same per-round marginals, no cross-round coupling — does not contain. SHA-256 is the right place to test it because it is a perfectly rigid synthetic environment in which the ground truth is known, so the claim that the residue is readable can be verified before it is believed of a system, such as a protein, where it cannot be checked. The engine implementation was verified against the standard SHA-256 test vector throughout.

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2. Method — Reading the Residue as Coupling

The measurement compares the real trajectory against a matched null. For each of several thousand random messages, the sixty-four per-round working states are recorded and reduced to a low-dimensional per-round symbol; the mutual information between the symbol at round t and the symbol at round $t+k$ is estimated with the Miller–Madow bias correction and averaged over all valid round pairs, for lags k from one to twelve. The null is generated by independently permuting each round's symbol across the message axis, which preserves every per-round marginal distribution exactly while destroying all coupling between rounds; the null band is the mean and standard deviation of this statistic over many such permutations. A real value above the null band at a given lag means the trajectory carries cross-round structure that the marginals cannot account for. This is the residue made measurable: not a trajectory read off a relaxation, but a constraint read off a coupling.

The choice of readout is the experiment's hinge, and the paper deliberately runs several. A parity readout reduces each round's state to the eight single-bit parities of its working registers — the maximally coarse GF(2) shadow, marginally near-uniform, and the natural first probe. Because parity discards exactly the carry information, and carries are where this construction displaces tension, the parity result alone cannot distinguish a property of the construction from a limitation of the lens. The probe is therefore exhausted with three further readouts: the low-order bits of the freshest injection registers, an injection-site state shadow formed from the low byte of the exclusive-or of the two newly written registers, and an explicit carry-bit readout that records the carry-out bits of the round's three structural modular additions. If the structure is a recoverable trajectory, a richer state readout should reveal more of it; if the structure is in the displaced tension, the carry readout should see what parity cannot.

3. Result — The Coupling Is Real and Bounded

Under the parity readout the residue is unmistakable and sharply bounded. Against a matched null whose per-round symbol entropy is 7.81 of a possible 8 bits — the marginals are near-maximally uniform, as a maximal-wash construction requires — the pairwise mutual information between rounds exceeds the null by 2.66 bits at lag one, by 1.07 bits at lag two, and by 0.29 bits at lag three, then collapses to within the null band at lag four and remains there for every subsequent lag. The structure is not in the marginals, which are featureless; it is in the coupling, and the coupling has a horizon of exactly three rounds. This horizon is the round function's local diffusion depth — the number of rounds over which the message schedule and the working-register pipeline fully mix — and it is the same period-three structure the program has encountered repeatedly across its substrate and constant-field work.

Lag k	$I(t;t+k)$ real (bits)	$I(t;t+k)$ null (bits)	Excess (bits)	z
1	5.92	3.26	2.66	3234
2	4.34	3.26	1.07	1094
3	3.55	3.26	0.29	283

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4	3.26	3.26	0.001	1.3
5–12	≈ 3.25	≈ 3.25	≈ 0	< 3

Table 1. Engine 28/29 — parity readout. Cross-round mutual information against a matched null: a three-round envelope, zero beyond. Marginals (Engine 27) are featureless; the structure is entirely in the coupling.

4. Exhausting the Probe — Two Channels, Two Depths

A three-round bound is only publishable once the lens is ruled out, so the probe is exhausted. The result is not a simple widening of the horizon; it is a discovery that the residue is structured by channel. The two richer state readouts — the low-order bits of the injection registers and the injection-site state shadow — show no cross-round coupling above the matched null at any lag. This is the control that settles the interpretation: if the residue were a recoverable trajectory, a richer view of the state would reveal more of it, and instead it reveals none. The structure was never in the state marginal at any resolution; a sharper picture of the state shows nothing, because the displaced tension is not in the state, it is in how the tension moves through the construction.

The explicit carry readout sees what parity and the state shadows cannot, and what it sees is a second, distinct structure at a different depth. The carry channel is silent at lags one, two, and three — the very lags where the parity envelope is loudest — and then produces a single sharp echo at lag four, exceeding the matched null by 0.12 bits with a z of 366, before falling silent again at every subsequent lag. This is not the parity envelope reaching further; it is a separate coupling in a separate channel, a clean delayed echo where the parity channel has a smooth local decay. The residue is therefore not one horizon but two distinct couplings: a parity envelope decaying over lags one to three, and a carry echo firing once at lag four.

Readout	Per-round entropy	Cross-round structure	Horizon
Parity (8 register parities)	7.81 bits	envelope, lags 1–3	3 rounds
Low bits of (a,e)	3.97 bits	none above null	0
$(a \oplus e)$ low byte	7.95 bits	none above null	0
Carry-out bits	3.82 bits	single echo at lag 4	lag-4 spike

Table 2. Engine 29 — probe exhausted. Richer state shadows show no coupling (killing the trajectory reading); the carry channel shows a distinct lag-4 echo. The residue is channel- and depth-structured.

5. Sourcing the Echo — the Pipeline, Not the Schedule

An unexplained spike is a reviewer's first question, and the program's discipline is to publish no number it cannot source, so the lag-four carry echo was tested head to head against its two candidate mechanisms. The first candidate is the message schedule: from round sixteen onward each message word is computed by a recurrence on earlier words, which could couple carries at a fixed lag. The second candidate is the register pipeline itself: the round function shifts its eight working registers by one slot per round, so a carry absorbed at one structural addition re-enters an active addition a fixed number of rounds downstream, independent of how the message words are produced. The two mechanisms make a clean, opposite prediction in the first sixteen rounds, where the message words are the raw input and the schedule recurrence is inactive. If the echo is the schedule, it must be absent in those pre-recurrence rounds and appear only from round sixteen. If it is the pipeline, it must be present from round zero, uniformly.

The measurement is decisive. The lag-four carry echo is present with equal strength in both regimes: in the recurrence block it exceeds the null by 0.117 bits, and in the pre-recurrence block, where the schedule is merely the raw message and contributes no coupling, it exceeds the null by 0.138 bits — if anything slightly stronger. A per-starting-round scan confirms the echo fires at every lag-four pair from round zero through round fifty-nine, with no discontinuity at the round-sixteen boundary where the schedule recurrence switches on. The schedule is ruled out. The echo is the register-pipeline latency of the round function's own eight-slot shift — a structural property of the construction's internal wiring, present whether the message words are scheduled or raw. This is the stronger result, because it ties the carry residue's depth to the same register geometry the program has been mapping throughout, rather than to an incidental feature of the schedule.

Round regime	lag-4 carry excess	z	Schedule active?
Recurrence block (16–63)	0.117 bits	557	yes
Pre-recurrence block (0–15)	0.138 bits	406	no

Table 3. Engine 29B — the lag-4 echo is present with full strength in the pre-recurrence block where the schedule is inactive. Mechanism: the register-pipeline latency, not the message schedule.

6. What the Residue Is, and What It Is Not

Assembled, the three engines give a precise and bounded picture of the residue, and it differs from both intuitions that preceded it. The residue is not a fog that fades smoothly with distance: the carry channel is silent for three rounds and then speaks once at the fourth, which no monotone-decay model predicts. And the residue is not a recoverable trajectory that can be read off a relaxation: the richest state readouts show no coupling at any lag, so there is no movie of the history waiting to be played backward. The residue is two distinct constraints in two distinct channels at two distinct depths — a parity envelope carrying the local diffusion mixing over three rounds, and a carry echo carrying the register-pipeline latency at lag four — each

invisible to marginals, each invisible to richer state shadows, each now mechanistically sourced. The field remembers different things for different distances depending on which thread of tension is read.

This is the constraint-priority thesis made quantitative in a way that is new for the program. The displaced tension is not uniform; it is organized by channel, with a measured range and a known mechanism for each. That confirms the central prediction — a maximal-wash construction hides recoverable structure in coupling, not in marginals — and the confirmation is strong: 2.66 bits at the first parity lag and a clean 0.12-bit carry echo, both against matched nulls that contain neither. But the same measurement bounds the structure tightly. The recoverable information is narrow and short-ranged: a few bits, decaying over three rounds in one channel and a single delayed bit-fraction in another. It is a prune on the compatible set, not an inversion of the map. The program's conserved seam-parity obstruction is untouched by anything measured here — nothing recovers a unique history; what is recovered is a set of constraints at known channels and depths.

This is also the measured correction to a proposal that arose in the program's cross-model discussion: that one might relax such a construction at its terminal state and watch the field snap back along the unique history of its own curve, reading the seed from the relaxation. The correlation horizon is not an inversion horizon, and the readouts settle the matter directly. There is no channel in which the residue carries the global trajectory; the richest state shadows carry nothing across rounds, and the two channels that do carry structure carry only a three-round envelope and a single lag-four echo. A construction may be relaxed, but what is legible in the relaxation is the local constraint within these narrow windows, not the history beyond them. The residue reads the constraint; it does not retrace the curve.

7. Consequences for Fold Tomography

Read as the front end of a tomography that aims to read the history of a curve in the tension field — a protein's fold, a constant's structure, a digest's seed — these results define the instrument rather than abolishing it. The instrument has a verified signal: against a known knot, with ground truth available, the residue is readable, and its readability has been demonstrated, not assumed. But the instrument has a measured aperture, and the aperture is channel-specific. There is no single uniform window; there is a parity channel that constrains the immediate three-round neighborhood and a carry channel that pins a specific bit four rounds back. A reverse procedure built on this is not a crank that turns the construction back one clean notch at a time, and it is not a catapult that releases the whole curve at once. It is a multi-channel constraint solver: read the parity envelope and the carry echo at the terminal window, narrow the set of compatible local histories, step back within the aperture, and re-read — with the explicit understanding that each step leaves residual multiplicity, that those multiplicities compound backward rather than cancel, and that the conserved obstruction bars half of all local histories from unique resolution. The honest instrument is a bounded constraint reader with a known, narrow, channel-structured aperture. Validated on a known knot, it is worth something; pointed at a curve that cannot be independently checked, it must carry its aperture with it.

8. Limitations and the Next Engine

The measurements are pairwise. The parity envelope and the carry echo are both two-round couplings, $I(t; t+k)$, and a deterministic construction can in principle hide structure in higher-order dependence — triples $I(t; t+1; t+2)$ beyond what the pairwise sum captures — that no pairwise probe, real or null, can see. That higher-order question is the next engine, and it is the genuinely open one: whether the three-round parity envelope is exhausted by its pairwise content or carries an irreducible triple. The readouts here, though several, are not exhaustive; a readout tuned to a different internal channel could in principle reveal a third structure at a third depth, and the claim is bounded accordingly — what is shown is that the residue is channel-structured and that two of its channels are sourced, not that exactly two channels exist. The synthetic environment is the strength and the limit of this paper: SHA-256 makes the residue checkable against ground truth, which is why the measurement is trustworthy, but the bridge to a biological or mathematical curve, where the same cross-round coupling measurement would be applied to a system without a known seed, is argued from the shared ontology and not yet measured. Each of these is a place where the instrument has been calibrated and not yet pointed; none is a place where it has failed.

9. Conclusion

A construction engineered to make every readout featureless does not destroy its recoverable structure; it displaces it into coupling, and the coupling is real, sharp, bounded, and sourced. In the perfectly rigid synthetic environment where the answer can be checked, the residue of SHA-256 carries 2.66 bits of cross-round structure at the first parity lag and a clean delayed carry echo at lag four, both invisible to every marginal and to richer state shadows, the first a three-round envelope set by the round function's diffusion depth and the second a single echo set by its register-pipeline latency. The residue is not a fog and not a trajectory; it is channel- and depth-structured constraint, with a measured range and a known mechanism for each channel. This confirms what constraint-priority predicts — that the wash hides structure by balancing it into coupling rather than by destroying it — and it bounds that structure precisely, correcting with measurement the hope that a construction can be relaxed and its full history retraced. What looks like randomness is constraint displaced into coupling too narrow and too channel-specific to read from a single aperture. The residue reads the constraint, within a known window and not one step beyond it. The next engine asks whether that window hides an irreducible higher-order knot, or whether its content is exhausted by the pairwise couplings measured here.

Acknowledgments and Reproduction

Engines 28, 29, and 29B were designed in dialogue and executed live by Claude (Anthropic) under the author's NEXUS constraints and session standards; every number in this paper is taken verbatim from those seeded runs. The residue-trace engine (`engine28_residue.py`), the carry-exhaust probe-upgrade engine (`engine29_carry.py`), and the mechanism-test engine (`engine29b_mechanism.py`) accompany this document for independent reproduction, each with the SHA-256 implementation check against the 'abc' test vector. The matched-null methodology — per-round permutation preserving marginals and destroying coupling, with

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Miller–Madow-corrected mutual information — is shared across all three. This paper builds directly on the constraint-priority synthesis and the marginal results of the canonical-carrier engine (Engine 27).

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